Interview with Maurizio Dècina

Good morning, Professor Dècina. Thank you for agreeing to meet with us.

You are now in an institution where information technologies are taught: the Politecnico di Milano. Information typically originates from electronics, the "parent subject" that enables the technical processing of information.

Historically, electronics has been divided into three branches: telecommunications, automatic control, and computing. Automatic control gave rise to two important fields: bioengineering and management engineering.

Today, the degree in management engineering is going through a period of great success; it is a course of study that also provides notions of economics, technology, and other knowledge relevant to working in a company. Keep in mind that today, in our country, there is great demand for information engineers, amounting to almost half a million positions. Indeed, those who graduate in engineering from the Politecnico di Milano generally receive many job offers, particularly nowadays, as technology develops so quickly in many sectors that it requires an ever-greater presence of experts. It is estimated that within a few decades the demand for engineers will be even greater than it is today.

Who was Professor Francesco Carassa to you and to the Politecnico di Milano?

Francesco Carassa was "the master" for all of us who worked in telecommunications, just as Professor Emanuele Biondi was for automatic control and Professor Luigi Dadda for computing. Francesco Carassa was the great pioneer of telecommunications studies in our country: he dreamed of using radio frequencies with extremely high capacity. As you know, there is the hertz, corresponding to one cycle per second; the kilohertz, corresponding to a thousand cycles; the megahertz, to a million; the gigahertz, to a billion; and the terahertz, to a trillion. Today we have almost reached the terahertz. Francesco Carassa, however, who was born in 1922, in his youth tended to use microwaves, that is, the gigahertz range. To do this, he in turn had a great mentor, the mentor of all those in our country who worked and still work in electronics, Professor Francesco Vecchiacchi.

Professor Vecchiacchi was an industrialist: in his time, Italy was an important country from an industrial standpoint, unlike today. Carassa worked with Vecchiacchi at Magneti Marelli. When Vecchiacchi died in 1955, Francesco Carassa became director of research at Magneti Marelli, a role he held until 1962.

The role of Magneti Marelli in the 1950s was fundamental. Television had arrived in Italy in the first half of the 1950s. In the second half of the 1950s, it became necessary to study how television worked: a tower emits a signal received by everyone, but this tower has a limited range — a single tower cannot transmit the signal across all of Italy, so towers had to be built in every major Italian city. And what did Francesco Carassa do? He used horn radio antennas with parabolic reflectors to build a network of radio links covering the whole of Italy. From 1955 to 1960, Marelli built 1 GHz microwave RadioLink connections for RAI, to allow the television channel to be broadcast throughout the country. We can therefore say that Carassa contributed to Italy's significant cultural growth, given how important television was for literacy in various parts of our country.

In 1962, Carassa decided to move into academia. That very year he won a professorship at the Politecnico di Milano and began to dream of working with higher frequencies. At the time, satellites used the highest frequencies, such as 40 GHz, because they were mainly geostationary and therefore transmitted well at these frequencies. Carassa worked on the Italian satellite project, Sirio, starting in 1969. It took eight years to launch it into geostationary orbit, since it weighed 398 kg. Being geostationary, Sirio remained fixed at zenith, meaning it rotated in sync with the Earth. It was used for measurement, telephony, and meteorology experiments. Initially it covered Europe and America; later, in its last two years, its position was changed so that it covered Asia.

Toward the end of the 1970s, the Politecnico equipped itself with a satellite station, used to manage all the work related to the Italian satellite Italsat, to which Carassa devoted much time and effort.

Carassa, a man professionally born in industry, brought all his knowledge into the university and founded the field of telecommunications. He was rector of the Politecnico between 1969 and 1972. He was President of the Council of the European Space Agency. He was one of the fathers of the Italsat satellite. He launched two consortia within the Politecnico di Milano (a consortium being an entity funded by companies that operates within the Politecnico):

Cefriel, dedicated to Information & Communication Technology applications, and CoreCom, dedicated to optical communications. In this way he brought industrial technology into the Politecnico. He was President of CSELT (a telecommunications research institute in Turin), President of Italtel, and of numerous other scientific institutions.

In 1983, Carassa won the most prestigious award in telecommunications, the Marconi International Fellowship, for his contributions in the field of satellite communications.

Professor Dècina, could you tell us about your career path?

I studied engineering in Rome in the 1960s, graduated in 1966, and began working in Rome. During this period my mentor was Professor Peroni, one of Vecchiacchi's students and collaborators. In 1980, at age 37, I won a professorship at the University of Rome and then went to the United States, to Bell Laboratories, to study the operation of switching machines, that is, telephone exchanges. I later worked at Italtel, where I served as Director of Industrial Research. In that role, for four years I collaborated with a thousand other people to build a single piece of software to run the machines. In 1986, Carassa helped move my professorship from Rome to Milan. In 1988 I founded Cefriel together with Carassa.

Throughout my career I have worked on the internet. The internet has evolved in a very straightforward way. Initially there was the internet of the web, the one we use every time we search with a browser; then came a second phase of the internet, which began at the start of this century, that of social network internet; finally there is a third generation of the web, the Internet of Things (IoT). This latest phase consists of sensors placed everywhere to monitor or measure various situations relevant to automation. IoT breaks down into two applications: Massive IoT and Mission Critical IoT, in which human life is at stake. An example of the latter application is connected cars, which consist of a single automobile used by multiple people, since thanks to self-driving it moves on its own between various locations according to customers' needs. A connected car must have sensors inside it that allow it to drive safely, without endangering the health of passengers or pedestrians.

Mission Critical Internet goes hand in hand with human life; for example, it made possible the first remote surgical operation via a device that sends signals into the operating room. In this situation, immediate connectivity is a truly important safety aspect, since the reaction time is on the order of milliseconds, and the network allows for rapid feedback signals.

The Internet of massive things covers everything related to the environment: it can be used, for example, to help prevent landslides, earthquakes, and floods, and it is applied in real time.

Low latency is also essential for entertainment. With mass online entertainment, networked gaming applications need minimal delay to allow for immediate gameplay.

In 2015 I turned my attention to cellular systems. Bits are the elementary units of information.

First-generation mobile phones had a capacity of ten thousand bits per second. By 2010 we had reached the fourth generation, with a capacity of 100 megabits per second. By 2020 we would reach several gigabits per second.

Bits are used not only by cellular radio technology but also by Wi-Fi technology, which, covering a smaller area, has greater capacity. Optical fibers have a capacity of several terabytes per second.

Within the Internet of massive things there are entire technological cities: smart cities.

The 5G system marks a great revolution and is highly innovative; transmission increases, and data centers are located even a thousand kilometers away. With cloud computing, hardware is no longer owned individually; the operating and application system is shared. Large companies are pushing in this direction: Amazon and Google. The purpose of this is to build memory, store and process information, and finally transmit it through communication. Cloud software is placed even a thousand kilometers away because speed is extremely high. With virtualization there is a single operating system with containers equipped with software containing microsystems that are easy to manage, test, and deploy. DevOps is the great orchestration system that organizes and optimizes all applications. We have seen how, in the 5G system, Mission Critical applications include autonomous vehicles that enable telemedicine and robotics.

Viruses constitute a danger, and indeed cybersecurity requirements are very high. Hacking must be strictly controlled so as not to compromise either the confidentiality or the ownership of so-called "data ownership."

How would you describe your work as a teacher?

I would describe my career as a professor as unusual compared to others' paths. I've had experiences of various kinds in different parts of the world. I worked as a researcher, in industry, as a Full Professor in Rome, as Director of CEFRIEL (Research and Training Center), and since 1987 I have collaborated with Professor Carassa. I retired in 2012. I served as a commissioner of AGCOM and as President of INFRATEL, the company responsible for developing optical fiber in market-failure areas. Optical fiber is essential for the internet of the future.

These are my honors: in 1997 I received the International Communication award, the same one Francesco Carassa had received in 1992 — Professor Carassa for his work in the field of satellites, and I for my work at the American Institute of Electrical and Electronics Engineers. I also had the honor of serving as President of the Communications Society.

Why, in your opinion, should we enroll in the engineering faculty?

Become engineers, because Italy needs them: we are short a million engineers. Engineering studies should also be considered by young women: when I worked at ITALTEL, my boss was a woman, Marisa Bellisario, CEO of ITALTEL, and I was a professor who had just returned from the United States working for her. Women are excellent at engineering because they have two qualities characteristic of this profession: they are meticulous and tenacious.

The shortage of engineers was already evident in my day: when I graduated in 1966, I received twelve job offers. In the end, I chose to become a research fellow because my father wanted me to become a professor.

In my life I have also been an entrepreneur: I went to the United States, where I learned the basics of the internet, and once back in Italy I invested all my savings in a small internet company. All the savings I invested in 2002 multiplied two hundredfold. I founded other companies, and some of them also failed, but what matters is getting things done. Engineers get things done.

Having industrial experience is very useful for an engineer. Collaborating with industry is fundamental: it contributes to career development and keeps one grounded in reality; it also develops skills that cannot be acquired at university, such as organizational skills, the ability to meet deadlines, and the ability to distinguish what is important from what is less important. As Pasteur said, discovery favors the prepared mind: it is essential to match work in a company, whose aim is to create products, with basic research, whose aim is to discover algorithms.

I taught telecommunications and security for twenty years at the Politecnico di Milano; I taught how to protect networks. Today there are eight billion phones in the world, and they are destined to become ever more numerous and ever more powerful. Cellular networks all talk to each other, thanks to the combined internet and telephone system. The evolution of 5G and 6G is set to continue over time: from the 1990s to 2020, researchers worked hard and well, and they will surely continue to work well from 2020 to 2040.

How does a hacker get into an account?

There are many types of hackers: for example, there are hackers who do security work.

Hackers must penetrate a perimeter: for instance, a company has a cyber perimeter, a cybersecurity boundary, where it keeps its files, and this perimeter must be protected. The hacker enters the network and begins exploring its features, finds a vulnerability, enters the internal network, and penetrates the company's network. Once there, they take control of a terminal and listen in, then steal the network administrator's passwords and take over the network.

To defend against this, it is necessary to ensure there are no vulnerabilities within the network and to prevent hacker attacks: this is done using artificial intelligence.

Within the internet there are various realities: 70% of the internet is private, belonging to Google, Amazon, and Facebook, each with its own networks made up of data centers and proprietary fiber optics — this reality is called the Private Internet; then there is the public internet (27%) and the dark web (3%). The dark web is often accessed anonymously, and the most horrendous things happen within it.

In these fields, things move extremely fast: Huawei, for example, became a world leader in telecommunications in just 12 years.

The ability to guarantee privacy, secrecy, and data ownership is what is needed today to be hired: all companies need these three characteristics to protect their data.